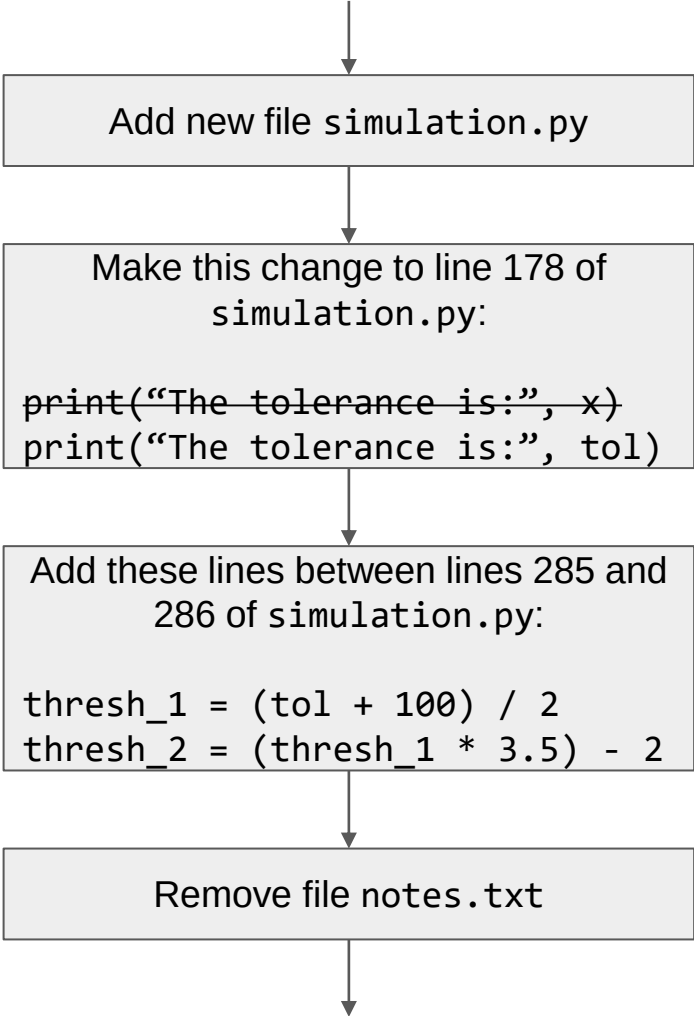


Git/GitHub Fundamentals

Git

Git is a *version control system* that makes it easy to track changes in the source code of a project, and also facilitates collaboration between multiple developers.



```
graph TD; A[ ] --> B[Add new file simulation.py]; B --> C[Make this change to line 178 of simulation.py:  
print("The tolerance is:", x)  
print("The tolerance is:", tol)]; C --> D[Add these lines between lines 285 and 286 of simulation.py:  
thresh_1 = (tol + 100) / 2  
thresh_2 = (thresh_1 * 3.5) - 2]; D --> E[Remove file notes.txt]; E --> F[ ]
```

Add new file simulation.py

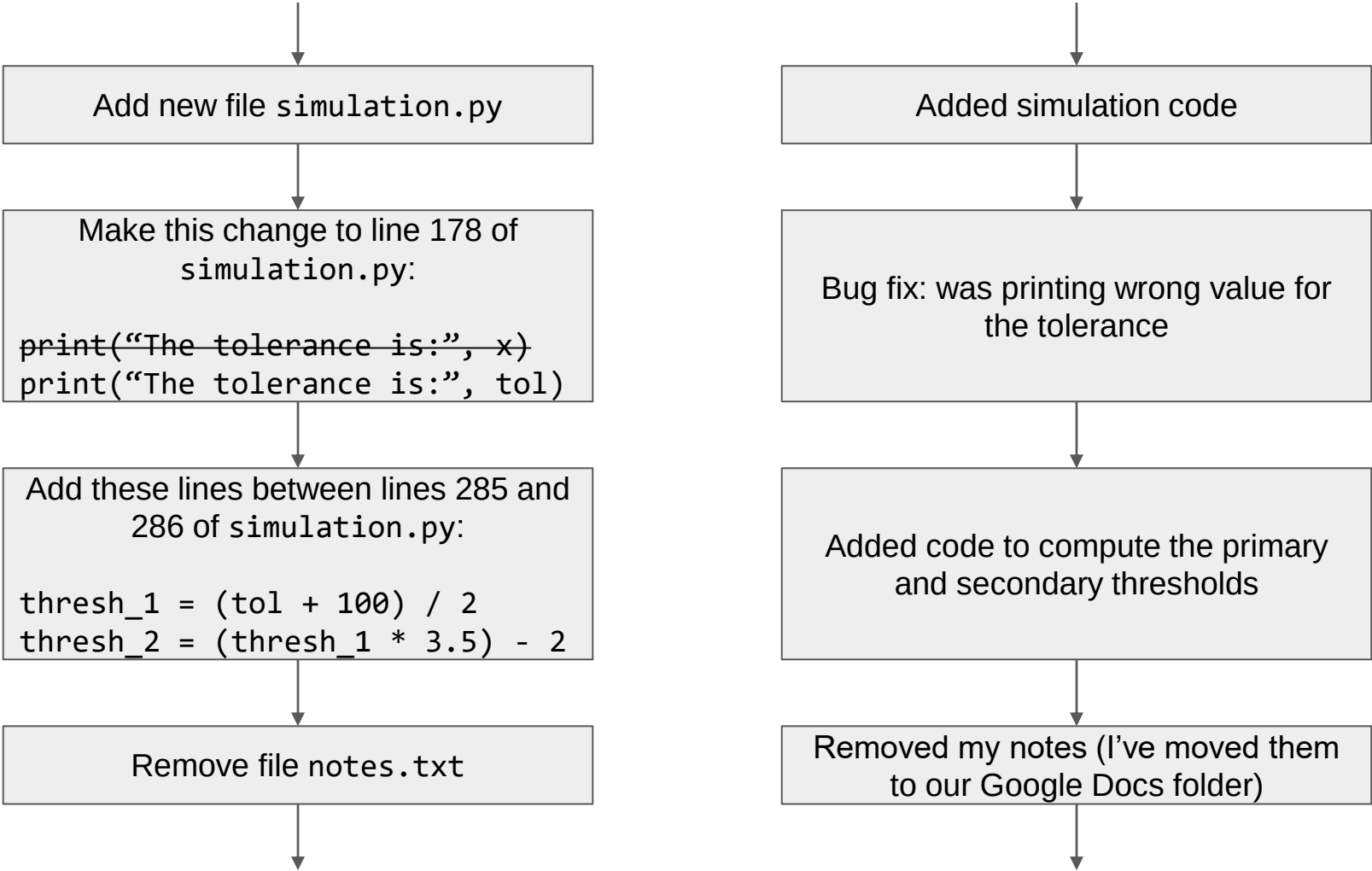
Make this change to line 178 of simulation.py:

```
print("The tolerance is:", x)  
print("The tolerance is:", tol)
```

Add these lines between lines 285 and 286 of simulation.py:

```
thresh_1 = (tol + 100) / 2  
thresh_2 = (thresh_1 * 3.5) - 2
```

Remove file notes.txt



```
graph TD; A[ ] --> B[Add new file simulation.py]; B --> C[Make this change to line 178 of simulation.py:  
print("The tolerance is:", x)  
print("The tolerance is:", tol)]; C --> D[Add these lines between lines 285 and 286 of simulation.py:  
thresh_1 = (tol + 100) / 2  
thresh_2 = (thresh_1 * 3.5) - 2]; D --> E[Remove file notes.txt]; E --> F[ ]; F --> G[Added simulation code]; G --> H[Bug fix: was printing wrong value for the tolerance]; H --> I[Added code to compute the primary and secondary thresholds]; I --> J[Removed my notes (I've moved them to our Google Docs folder)]; J --> K[ ]
```

Add new file `simulation.py`

Make this change to line 178 of `simulation.py`:

```
print("The tolerance is:", x)  
print("The tolerance is:", tol)
```

Add these lines between lines 285 and 286 of `simulation.py`:

```
thresh_1 = (tol + 100) / 2  
thresh_2 = (thresh_1 * 3.5) - 2
```

Remove file `notes.txt`

Added simulation code

Bug fix: was printing wrong value for the tolerance

Added code to compute the primary and secondary thresholds

Removed my notes (I've moved them to our Google Docs folder)

Git

- Any directory can be converted into a Git *repository*

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("hola!")  
print("hola, mundo!")  
print("hola, universo!")
```

Git Repository

Commits

```
$ git init
```

Latest Version

Staged for Next Commit

Untracked Files

hello.py

hola.py

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("hola!")  
print("hola, mundo!")  
print("hola, universo!")
```

Git Repository

Commits

Latest Version

```
$ git add hello.py hola.py
```

Staged for Next Commit

hello.py

hola.py

Untracked Files

hello.py

hola.py

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("hola!")  
print("hola, mundo!")  
print("hola, universo!")
```

Git Repository

Commits

Added Python files

- Added hello.py
- Added hola.py

Latest Version

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("hola!")  
print("hola, mundo!")  
print("hola, universo!")
```

```
$ git commit -m"Added Python files"
```

Staged for Next Commit

hello.py

hola.py

Untracked Files

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("Hola!")  
print("Hola, mundo!")  
print("Hola, universo!")
```

Git Repository

Commits

Added Python files

- Added hello.py
- Added hola.py

Latest Version

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("hola!")  
print("hola, mundo!")  
print("hola, universo!")
```

!

Staged for Next Commit

Untracked Files

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("Hola!")  
print("Hola, mundo!")  
print("Hola, universo!")
```

Git Repository

Commits

Added Python files

- Added hello.py
- Added hola.py

Latest Version

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("hola!")  
print("hola, mundo!")  
print("hola, universo!")
```



```
$ git add hola.py
```

Staged for Next Commit

hola.py

Untracked Files

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("Hola!")  
print("Hola, mundo!")  
print("Hola, universo!")
```

Git Repository

Commits

Added Python files

- Added hello.py
- Added hola.py

Forgot to capitalize in Spanish version

- Modified hola.py

Latest Version

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("hola!")  
print("hola, mundo!")  
print("hola, universo!")
```

```
$ git commit -m"Forgot to capitalize in Spanish version"
```

Staged for Next Commit

hola.py

Untracked Files

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")  
print("Hello, multiverse!")
```

hola.py

```
print("Hola!")  
print("Hola, mundo!")  
print("Hola, universo!")  
print("Hola, multiverso!")
```

bonjour.py

```
print("Bonjour!")  
print("Bonjour le monde!")  
print("Bonjour l'univers!")  
print("Bonjour multivers!")
```

Git Repository

Commits

Added Python files

- Added hello.py
- Added hola.py

Forgot to capitalize in Spanish version

- Modified hola.py

Latest Version

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("Hola!")  
print("Hola, mundo!")  
print("Hola, universo!")
```

Staged for Next Commit

Untracked Files

bonjour.py

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")  
print("Hello, multiverse!")
```

hola.py

```
print("Hola!")  
print("Hola, mundo!")  
print("Hola, universo!")  
print("Hola, multiverso!")
```

bonjour.py

```
print("Bonjour!")  
print("Bonjour le monde!")  
print("Bonjour l'univers!")  
print("Bonjour multivers!")
```

Git Repository

Commits

Added Python files

- Added hello.py
- Added hola.py

Forgot to capitalize in Spanish version

- Modified hola.py

Latest Version

hello.py

```
print("Hello!")  
print("Hello, world!")  
print("Hello, universe!")
```

hola.py

```
print("Hola!")  
print("Hola, mundo!")  
print("Hola, universo!")
```

```
$ git add hello.py bonjour.py
```

Staged for Next Commit

hello.py

bonjour.py

Untracked Files

bonjour.py

Filesystem

/home/username/my_code/hello/

hello.py

```
print("Hello!")
print("Hello, world!")
print("Hello, universe!")
print("Hello, multiverse!")
```

hola.py

```
print("Hola!")
print("Hola, mundo!")
print("Hola, universo!")
print("Hola, multiverso!")
```

bonjour.py

```
print("Bonjour!")
print("Bonjour le monde!")
print("Bonjour l'univers!")
print("Bonjour multivers!")
```

Git Repository

Commits

Added Python files

- Added hello.py
- Added hola.py

Forgot to capitalize in Spanish version

- Modified hola.py

More salutations

- Modified hello.py
- Added bonjour.py

```
$ git commit -m"More salutations"
```

Latest Version

hello.py

```
print("Hello!")
print("Hello, world!")
print("Hello, universe!")
```

hola.py

```
print("Hola!")
print("Hola, mundo!")
print("Hola, universo!")
```

bonjour.py

```
print("Bonjour!")
print("Bonjour le monde!")
print("Bonjour l'univers!")
print("Bonjour multivers!")
```

Staged for Next Commit

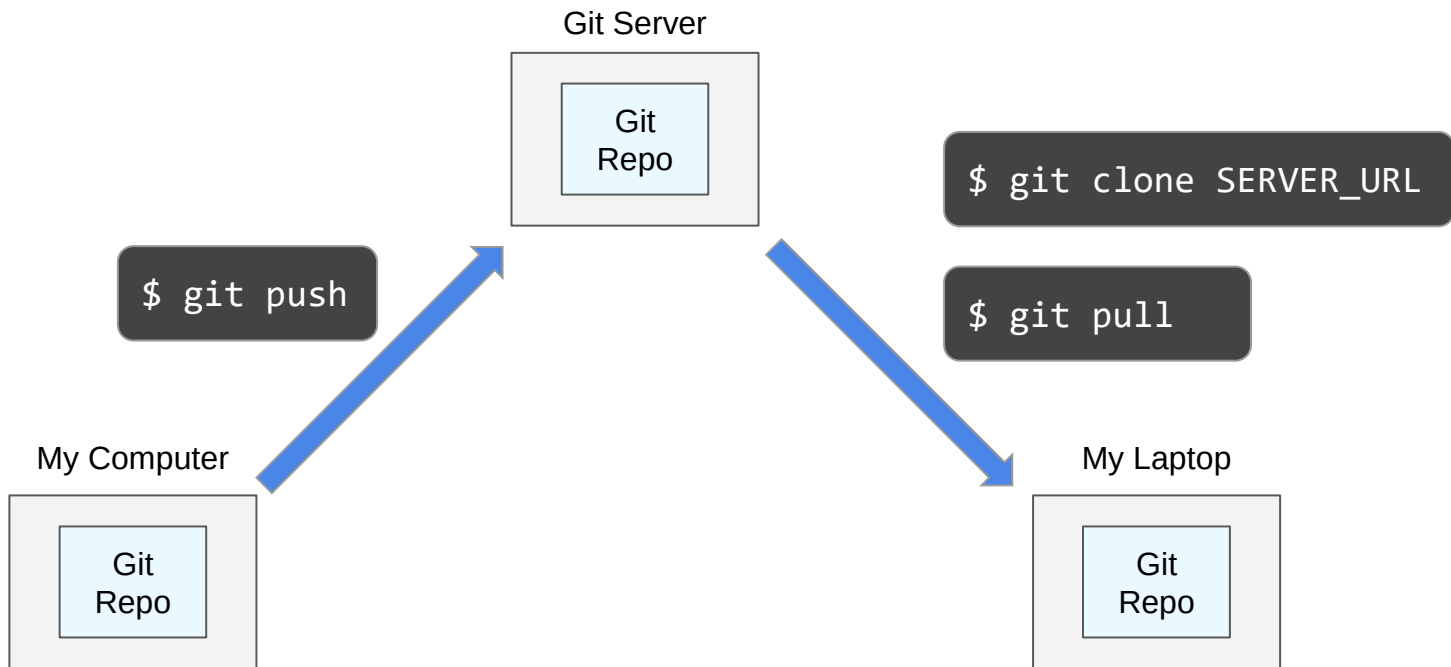
hello.py

bonjour.py

Untracked Files

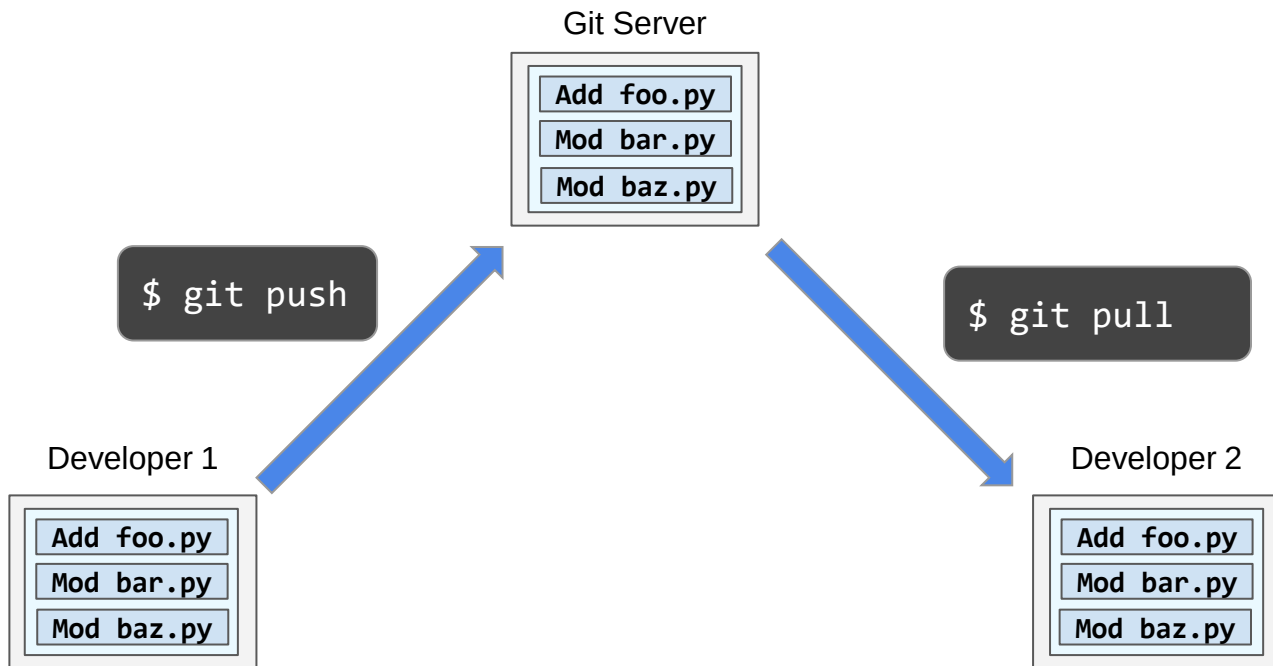
Git

Git also allows you to synchronize your repository with a server.



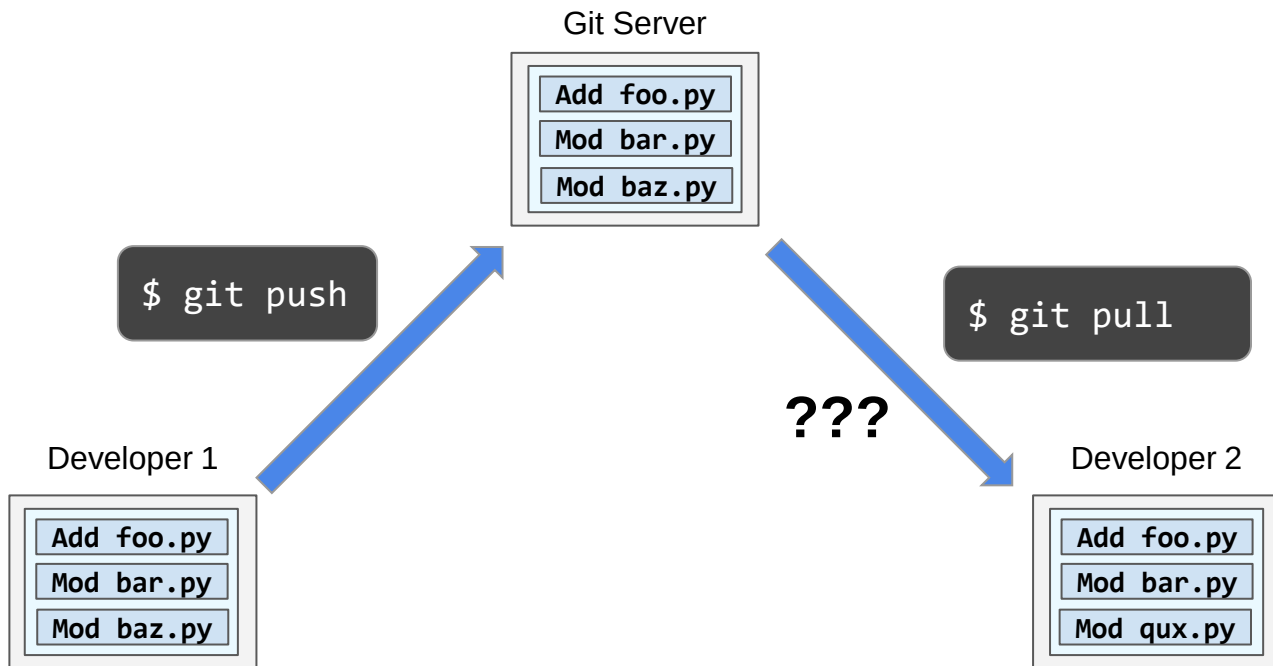
Git

This allows for collaboration between multiple developers



Git

Git provides a number of features for reconciling divergent changes



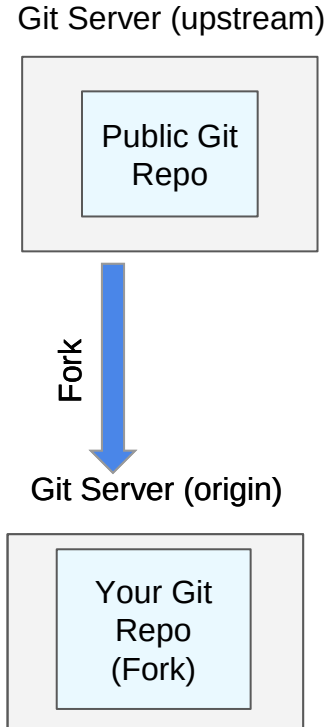
Demonstration

“Forking” a repository

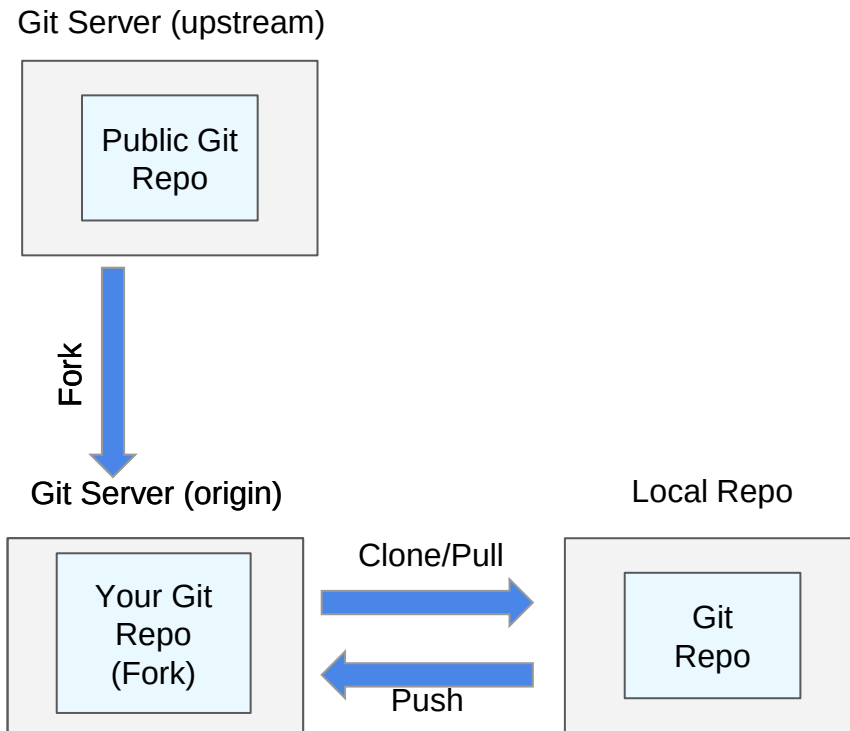
Git Server (upstream)



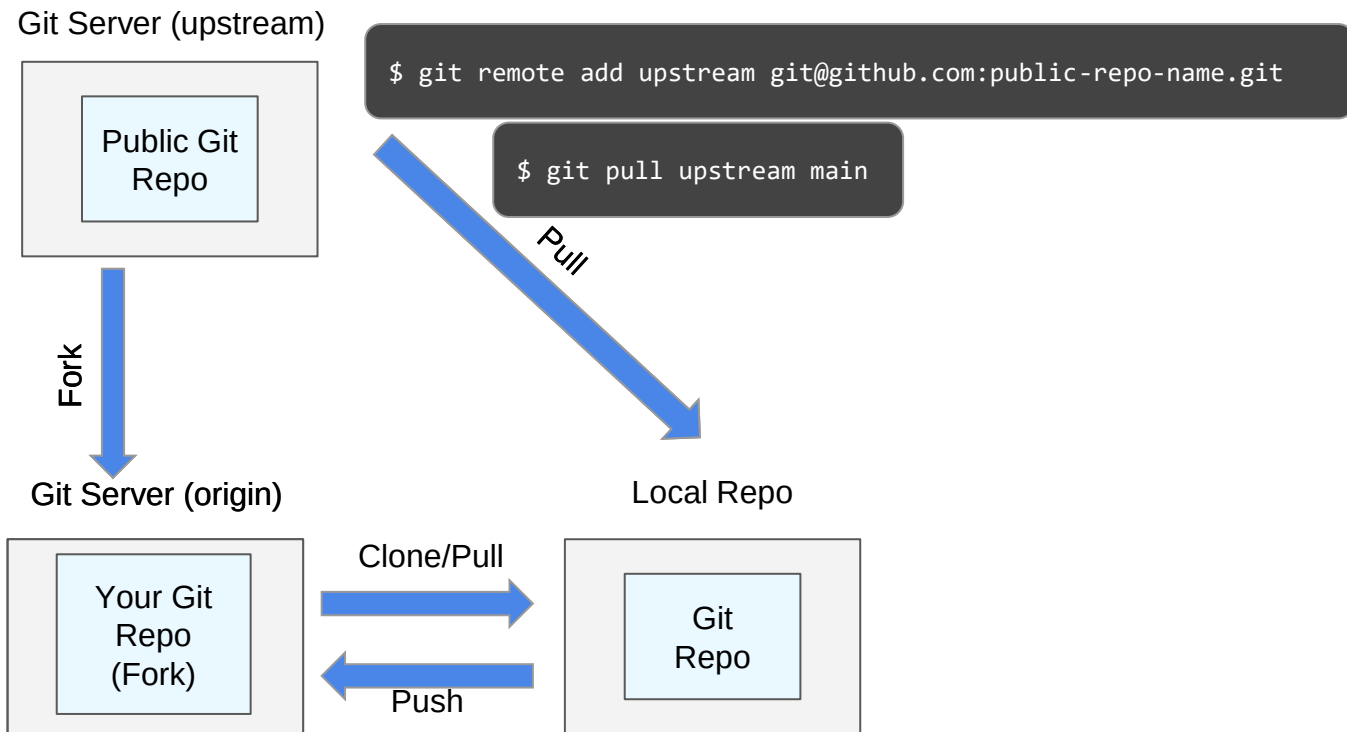
“Forking” a repository



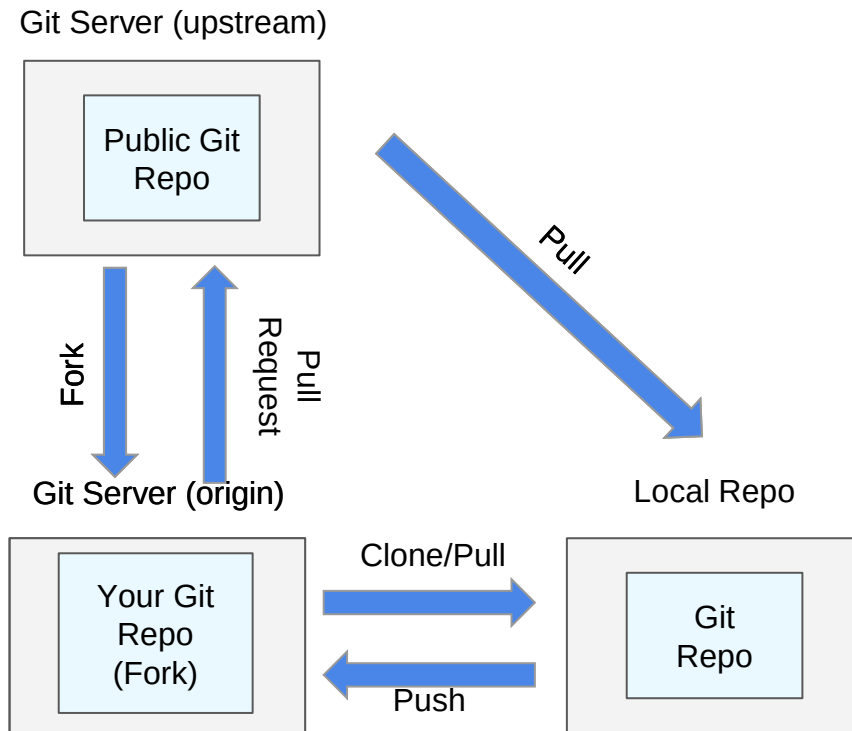
“Forking” a repository



“Forking” a repository



“Forking” a repository



Git/GitHub in this Class

- Programming Assignments: all of you will use separate private repositories for each assignment on GitHub
- Course Materials Repository: we will share files relevant for in-class activities with you here
 - To refresh this repository: `git pull`